

Final MVP Crop Catalog Fields & Location Comparisons

A practical schema for comparing sourced crop requirements with location-specific temperature, soil, water, region, and planting evidence across Texas.

How the evidence is divided

Dimension	Stored in crop catalog	Compared with
Temperature	Crop thermal ranges and thresholds	FortyGuard recent/local heat plus NASA long-term climatology
Soil	Crop texture, pH, and drainage requirements	USDA SSURGO map-unit properties, with optional soil-test override
Water	Demand, drought tolerance, and irrigation dependency	NASA precipitation, SSURGO water storage, and farmer irrigation status
Region and season	Texas applicability, planting window, maturity	Farm coordinates, region mapping, and proposed planting date
Evidence quality	Sources, confidence, review date, assumptions	Used by validation and confidence; not compared as agronomic conditions

Scope note: SSURGO estimates what the soil can store and how it drains. It does not report current soil moisture, rainfall, irrigation access, or legal water availability.

Comparison model

CropSage compares stable catalog requirements with normalized location evidence. The deterministic scorer produces separate suitability, risk, and confidence outputs; missing evidence lowers confidence rather than being treated as favorable.

Complete Catalog Field Matrix

All required crop-catalog fields are listed in one continuous table. The header repeats when the table crosses a page boundary.

Group	Crop catalog field	Location evidence	Comparison or purpose
Identity	crop_id	None	Stable internal identifier used across scoring, citations, and UI state.
Identity	common_name	None	Farmer-facing display name.
Identity	scientific_name	None	Prevents ambiguity between crops with similar common names.
Identity	production_use	Optional farmer objective	Separates production profiles such as grain corn and silage corn.
Identity	variety_scope	None	States whether requirements apply to a species, commercial type, or named cultivar.
Region	supported_texas_regions	Farm coordinates mapped to a Texas region	Determines whether the crop profile is supported at the selected location.
Region	regional_suitability	Resolved Texas region	Returns core, conditional, marginal, or unsupported.
Season	planting_windows_by_region	Region and proposed planting date	Checks whether the scenario date is inside a recommended planting window.

Group	Crop catalog field	Location evidence	Comparison or purpose
Season	days_to_maturity	Planting date and expected growing-season length	Checks whether the crop can mature before damaging cold or seasonal limits.
Temperature	optimal_temperature_range	NASA climatology and recent FortyGuard temperatures	Measures whether expected baseline and recent local temperatures are favorable.
Temperature	heat_stress_threshold	FortyGuard maximum temperatures	Detects crop-specific threshold exceedance.
Temperature	heat_exposure_duration	FortyGuard consecutive hourly observations	Separates a brief spike from persistent heat exposure.
Temperature	heat_sensitive_stages	Planting date, estimated crop stage, and FortyGuard time period	Applies stronger heat penalties during sensitive growth stages.
Temperature	frost_sensitivity	NASA historical minimum temperatures and seasonal timing	Identifies cold or frost risk for the proposed crop calendar.
Temperature	temperature_unit	Provider response units	Ensures all values are normalized before comparison.
Temperature	temperature_measurement_basis	FortyGuard and NASA parameter definitions	Prevents comparisons between incompatible temperature measurements.
Soil	preferred_soil_textures	USDA SSURGO soil texture	Measures texture compatibility for the selected point or farm polygon.
Soil	ph_tolerable_range	SSURGO representative pH or farmer soil test	Detects unsuitable acidity or alkalinity; a soil test may override mapped evidence.
Soil	drainage_requirement	USDA SSURGO drainage class	Detects drainage mismatch and potential waterlogging risk.
Water	water_demand	NASA growing-season precipitation, SSURGO water storage, and irrigation status	Classifies whether the location is likely to meet the crop's water demand.
Water	drought_tolerance	Calculated rainfall and soil-storage risk	Adjusts suitability when the location has a likely climatic water deficit.
Water	irrigation_requirement	Farmer irrigation availability	Identifies crops that depend on irrigation at the selected location.
Evidence	source_ids	None	Links each requirement to supporting publications or authoritative guidance.
Evidence	confidence	Provider quality and missing location evidence	Contributes to recommendation confidence; it is not crop suitability itself.
Evidence	last_reviewed	None	Records how recently the crop profile was checked.
Evidence	assumptions	Farm and scenario inputs	Discloses variety, regional, and data assumptions used in the result.
Administration	catalog_version	None	Supports versioned, reproducible catalog updates.
Administration	record_status	None	Marks the record as draft, reviewed, or approved.

Runtime Inputs and Interpretation

Required runtime location profile

Location input	Source
Latitude and longitude or field polygon	Map selection
Texas agricultural region	Derived from coordinates
Proposed planting date	Farmer scenario input
Temperature climatology	NASA POWER
Recent and near-term heat	FortyGuard
Historical growing-season precipitation	NASA POWER
Soil texture	USDA SSURGO
Soil pH	SSURGO; optional soil-test override
Soil drainage	USDA SSURGO
Available soil water storage	USDA SSURGO
Rainfed or irrigated	Farmer
Irrigation reliability	Farmer
Groundwater context, optional	Texas Water Development Board

MVP water-risk interpretation

Climatological rainfall + soil water-storage contribution + irrigation adjustment -> approximate crop water suitability or water risk. This is not an exact irrigation-volume calculation.

Optional later refinement: add gridMET reference evapotranspiration and a documented crop-water calculation. Texas Water Development Board data may add aquifer and nearby-well context, but it cannot prove farm-level irrigation access.

Missing-data rule: if a required location input is unavailable, mark that dimension unknown and reduce confidence. Do not silently assume favorable conditions.

Authoritative Data References

FortyGuard Temperature API documentation
<https://docs-api.fortyguard.com/docs>

NASA POWER climatology API
<https://power.larc.nasa.gov/docs/services/api/temporal/climatology/>

USDA NRCS SSURGO overview
<https://www.nrcs.usda.gov/resources/data-and-reports/soil-survey-geographic-database-ssurgo>

USDA Soil Data Access web services
<https://sdmdataaccess.sc.egov.usda.gov/webservicehelp.aspx>

gridMET dataset description
<https://www.climatologylab.org/gridmet.html>

Texas Water Development Board data services
<https://www.twdb.texas.gov/mapping/data-services.asp>

Interpretation boundaries

Evidence	Boundary
NASA POWER	Regional climate context; its meteorology grid is not field truth.
SSURGO	Mapped survey estimate; a laboratory soil test can supersede representative values.
TWDB	Regional groundwater context; not proof of ownership, pumping capacity, allocation, or access.
CropSage output	Suitability and risk decision support; not a yield prediction or irrigation prescription.

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